SWRT for nonlinear four-wave interactions in ecWAM

SWRT (sparse Webb–Resio–Tracy) computes the nonlinear four-wave interactions S_{nl} from K configurations of the WRT quadrature table, each a quadruplet and its mirror image, with one optimized coefficient per configuration. This patch adds SWRT to ecWAM as an alternative to the Discrete Interaction Approximation (DIA) in `SNONLIN`. In a 24-h global test, SWRT keeps H_s within 1 % of WRT, compared with 15 % for DIA, and takes 1.6 times the time per step of DIA.

The SWRT files and this guide are on the branch `swrt` of Jialunx/ecwam, a fork of ecmwf-ifs/ecwam (ecWAM 1.5.14).

1. Changes to ecWAM

The patch adds one source file and changes eight others. The default `INLSCHEME = 0` leaves ecWAM unchanged.

File	Change
`src/ecwam/snonlin_swrt.F90` (new)	`INISNONLIN_SWRT` reads the SWRT file and checks it against the spectral grid. `SNONLIN_SWRT` takes the arguments of `SNONLIN` and adds S_{nl} to `SL` and its diagonal term to `FLD`.
`src/ecwam/yowstat.F90`	Declares the switch `INLSCHEME`.
`src/ecwam/mpuserin.F90`	Reads `INLSCHEME` in namelist `NALINE`, 0 for DIA (default) and 1 for SWRT.
`src/ecwam/userin.F90`	Writes the scheme to `wam.log` and stops for other values.
`src/ecwam/initmdl.F90`	Calls `INISNONLIN_SWRT` when `INLSCHEME = 1`.
`src/ecwam/implsch.F90`	Calls `SNONLIN_SWRT` instead of `SNONLIN` in the source-term integration when `INLSCHEME = 1`.
`src/ecwam/wdfluxes.F90`	Calls `SNONLIN_SWRT` instead of `SNONLIN` for the wave-dependent fluxes at the initial time when `INLSCHEME = 1`.
`src/ecwam/CMakeLists.txt`	Adds the new file to the build.
`share/ecwam/scripts/ecwam_run_model.sh`	Passes `isnonlin`, `inlscheme` and `swrt_operator` of the yml settings to the namelist and the environment.

The SWRT files hold the deep-water interactions, which `ISNONLIN` scales in shallow water as it scales DIA. The diagonal term is the derivative of S_{nl} at a spectral bin with respect to the spectral density at that bin, taken analytically as in `SNONLIN`. Each quadruplet of SWRT moves action from k_3 to k_1, both on the spectral grid, at a rate proportional to $T = n_1\, n_3\, (n_4 - n_2) + n_2\, n_4\, (n_3 - n_1)$, where n_i is the action density at k_i and n_2 and n_4 are interpolated. Its diagonal term is proportional to $\partial T/\partial n_1$ at k_1 and to $-\partial T/\partial n_3$ at k_3, as in the WRT code Xnl (van Vledder 2006). SWRT adds diagonal terms only at k_1 and k_3, whereas DIA adds them also at the bins around its interpolated wavenumbers.

2. Installation

The branch can be cloned directly (video guide in the release swrt-video).

```
git clone -b swrt https://github.com/Jialunx/ecwam
```

An existing clone of ecWAM 1.5.14 takes the same changes from the patch.

```
git checkout 1.5.14
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/ecwam_swrt.patch
git apply ecwam_swrt.patch
```

ecWAM then builds as before and needs no new library. The SWRT files are in the folder `swrt/files` of the branch and go to any directory the model can read. The files for the other spectral grids of section 3 are in the same folder.

```
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/swrt_ec4x_K31L13.ww3op
curl -LO https://raw.githubusercontent.com/Jialunx/ecwam/swrt/swrt/files/wrt_table_K807.ww3op
```

SWRT file	Configurations	Offsets	Use
`swrt_ec4x_K31L13.ww3op`	31	13	runs with SWRT, selected for four times the S_{nl} cost of DIA
`wrt_table_K807.ww3op`	807	74	reference runs with WRT, all configurations with coefficient 1

The run in the video edits the yml file of the regression test `etopo1_oper_an_fc_048` of ecWAM to the settings below.

Setting	yml	Value
Spatial grid	`grid`	O48, global, about 1.9°
Directions	`directions`	36
Frequencies	`frequencies`	29 propagated, of NFRE = 36
First frequency	`fr1`, `ifre1`	0.0345 Hz, from fr1 = 4.177248E-02 Hz at ifre1 = 3
Time step	`timestep`	900 s for propagation and source terms
Physics	`iphys`	1
S_{nl}	`inlscheme`, `isnonlin`, `swrt_operator`	SWRT with `swrt_ec4x_K31L13.ww3op`, isnonlin = 0
Period	`analysis.begin`, `forecast.end`	12 UTC 31 December 2022 to 12 UTC 2 January 2023, 48 h
Forcing	`forcings`	ECMWF operational 10-m winds and sea ice, analyses every 6 h to 00 UTC 1 January 2023, then forecasts every hour
Validation	`validation`	removed, as it checks the results of DIA

3. Spectral grid

The SWRT files depend on the spectral grid only, not on the spatial grid or the time step. The spectral grid of ecWAM is set by the number of directions, the number NFRE of frequencies, and the reference frequency fr1 of bin ifre1, and its frequency ratio is fixed at 1.1. NFRE is the number of frequencies of the source terms, whereas `frequencies` in the yml file sets the number of propagated frequencies. The branch holds SWRT files for the common spectral grids below, each selected as the operational file is, for about four times the S_{nl} cost of DIA.

SWRT file	Directions × NFRE	First frequency (Hz)	Ratio
`swrt_ec4x_K31L13.ww3op`	36 × 36	0.0345	1.1
`swrt_d24_f0345_K27L16.ww3op`	24 × 36	0.0345	1.1
`swrt_d24_n30_f0345_K29L14.ww3op`	24 × 30	0.0345	1.1
`swrt_d24_n40_f0345_K27L16.ww3op`	24 × 40	0.0345	1.1
`swrt_d12_f0418_K29L14.ww3op`	12 × 36	0.0418	1.1
`swrt_d24_f0418_K27L16.ww3op`	24 × 36	0.0418	1.1
`swrt_d36_f055_K29L14.ww3op`	36 × 36	0.055	1.1

The first frequencies 0.0345, 0.0418 and 0.055 Hz correspond to fr1 = 4.177248E-02 Hz at ifre1 = 3, the same fr1 at ifre1 = 1, and fr1 = 0.055 Hz at ifre1 = 1. A run names the file of its spectral grid in `swrt_operator` (section 4), and the model stops with a message when the file and the grid differ. The files with 30 and 40 frequencies also need NFRE = 30 or 40 in namelist NALINE, which the standalone scripts set to 36. The scripts in `tools/` of the SWRT code archive (doi:10.5281/zenodo.23654016) build a file for any other spectral grid in about an hour.

4. Running

Setting	Where	Value
`INLSCHEME`	namelist `NALINE`	1 for SWRT, 0 for DIA (default)
`WAM_SWRT_OPERATOR`	environment	path of the SWRT file
`ISNONLIN`	namelist `NALINE`	shallow-water scaling 0, 1 or 2, as for DIA
`WAM_SWRT_DIAG`	environment	`0` switches the diagonal term off (optional)

The standalone scripts read the same settings from three lines of the yml file.

```
physics:
    inlscheme: 1
    isnonlin: 0
    swrt_operator: /abs/path/swrt_ec4x_K31L13.ww3op
```

The log file `wam.log` names the scheme, the SWRT file, and its numbers of configurations and offsets. The model stops with a message when the variable is unset or the file is missing.

5. Test results

5.1 Synthetic and ERA5 spectra

The error ε of the direction-integrated S_{nl} (van Vledder 2006, Eq. 58) is computed against WRT on 300 synthetic spectra that were not used in the selection, half of them single-peaked and half bimodal. SWRT follows WRT closely, whereas the error of DIA exceeds the magnitude of S_{nl} itself.

Method	Median ε (%)	95th percentile of ε (%)
SWRT with K = 31	8	18
DIA of ecWAM	125	190

ERA5 spectra exist only on the grid of ERA5, 24 directions and 30 frequencies, and are tested with the ERA5 file of section 3. Over 1171 deep-water ERA5 spectra on a 5° grid at 00 UTC 1 January 2019, the median ε is 8 % for SWRT and 72 % for DIA. In an example of a wind sea with a swell, with ε of 8 % for SWRT and 55 % for DIA, SWRT reproduces the transfer of WRT in frequency and direction, whereas DIA overestimates its largest value by a factor of 1.6 and adds transfer at directions where WRT has none (Fig. 1).

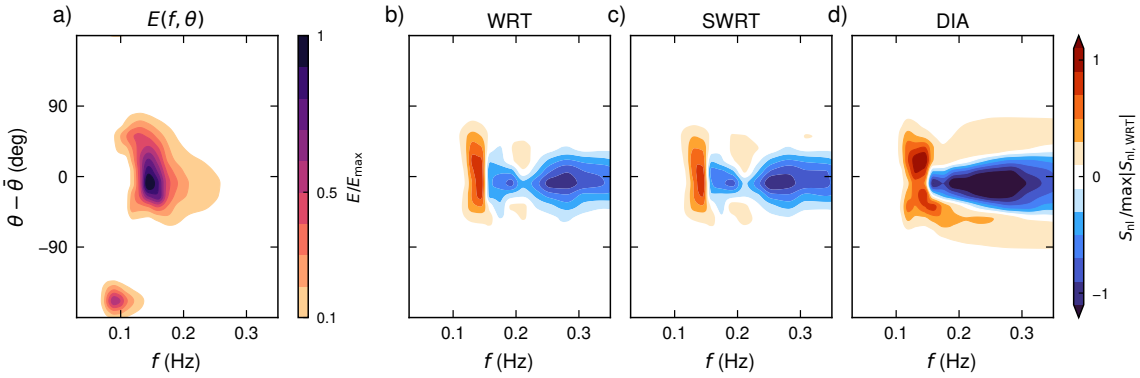


Fig. 1. An ERA5 spectrum of a wind sea with a swell at 35° S, 155° E, 00 UTC 1 January 2019. (a) E(f, θ) / E_{max}; (b)–(d) S_{nl}(f, θ) from WRT, SWRT and DIA, divided by the largest $|S_{nl}|$ of WRT. Directions are relative to the mean direction θ̄.

5.2 Global test

The global test extends the regression case `etopo1_oper_an_fc_048` of ecWAM to 24 h on the operational spectral grid of 36 × 36, of which 29 frequencies are propagated, from 12 UTC 31 December 2022 to 12 UTC 1 January 2023. It starts from rest with the ECMWF operational 10-m winds and sea ice, the physics of iphys = 1 and time steps of 900 s. The errors ε_{Hs} and ε_{Tz} are the relative differences from the run with WRT, summed over the hourly outputs from t = 1 h and over the sea points where H_s of WRT is at least 0.1 m.

Method	ε_{Hs} (%)	ε_{Tz} (%)	Mean H_s at 24 h (m)
WRT	–	–	1.20
SWRT with K = 31	0.7	0.8	1.19
DIA	15	9	1.35

SWRT keeps H_s within 1 % of WRT at every hour and nearly everywhere, while DIA gives 15 % higher values (Figs. 2 and 3).

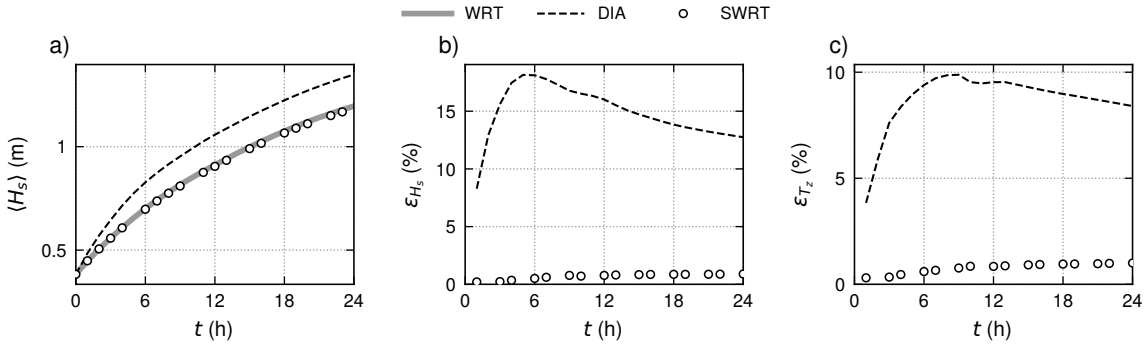


Fig. 2. Global statistics of the 24-h ecWAM test with S_{nl} from WRT (thick gray line), DIA (dashed) and SWRT with K = 31 (circles). (a) Mean H_s over the sea points; (b) ε_{Hs} and (c) ε_{Tz} at each hour.

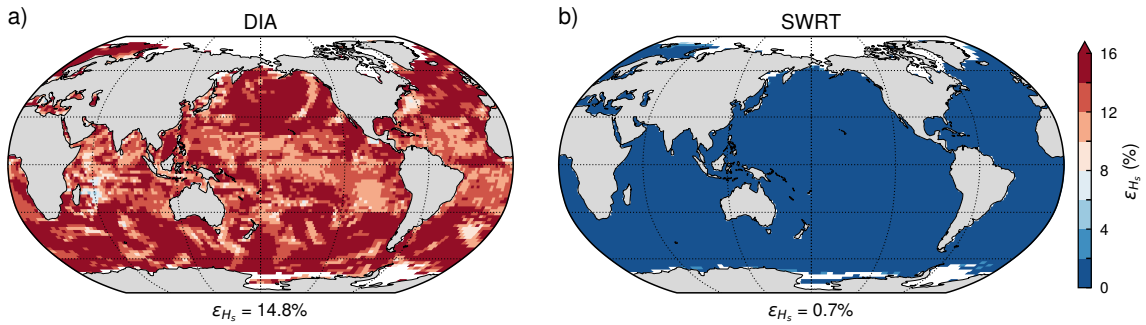


Fig. 3. ε_{Hs} at each sea point over the 24-h ecWAM test for (a) DIA and (b) SWRT with K = 31, with ε_{Hs} over all points under each map. Seas without a value are white, mostly the ice-covered polar seas.

5.3 Cost

The cost is measured with DR_HOOK in the optimized build of the 24-h runs, on one of four MPI tasks, and is given relative to DIA. WRT runs through the same vectorized routine as SWRT, and the difference in cost therefore comes from the numbers of configurations and offsets alone.

Method	S_{nl}	Time step	Run
SWRT with K = 31	4.0	1.6	1.4
WRT	68	14	11

S_{nl} takes about a fifth of a time step with DIA, half with SWRT and nearly all of it with WRT (Fig. 4).

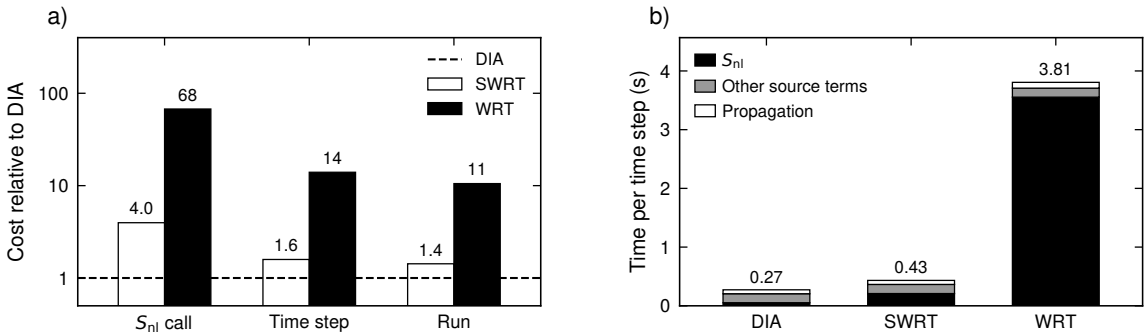


Fig. 4. Cost of S_{nl} in the 24-h ecWAM test. (a) Cost of SWRT with K = 31 (white) and WRT (black) relative to DIA (dashed line) for one S_{nl} call, one time step and the run; (b) time per time step, split into S_{nl}, the other source terms and propagation.